Home Reading 07 — Feedback

Before Class

Preview Lecture 7

Read Lecture 7: Feedback.

Textbook Reading

- Required: Ogata, System Dynamics, 4th ed., Section 10-2, pp. 494–498 (block diagrams and feedback simplification).
- Page numbers above are printed textbook pages. Use the Lecture slides for the remaining worked examples.

This Week Introduces

Closed-loop paths, stability, reference tracking, disturbance rejection, sensor effects and steady-state error.

Tutorial Preparation

Read Tutorial 7 before class. Be ready to:

- Q1: Derive reference and load-disturbance paths for a flywheel speed loop.
- Q2: Distinguish output disturbance, sensor bias and measured error.
- Q3: Read a feedback stability boundary from supplied evidence.

Check Your Understanding Before Class

Before class, try to answer these two questions:

- Why can negative feedback improve accuracy but still create instability?
- Why does the disturbance entry point change its transfer function to the output?

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